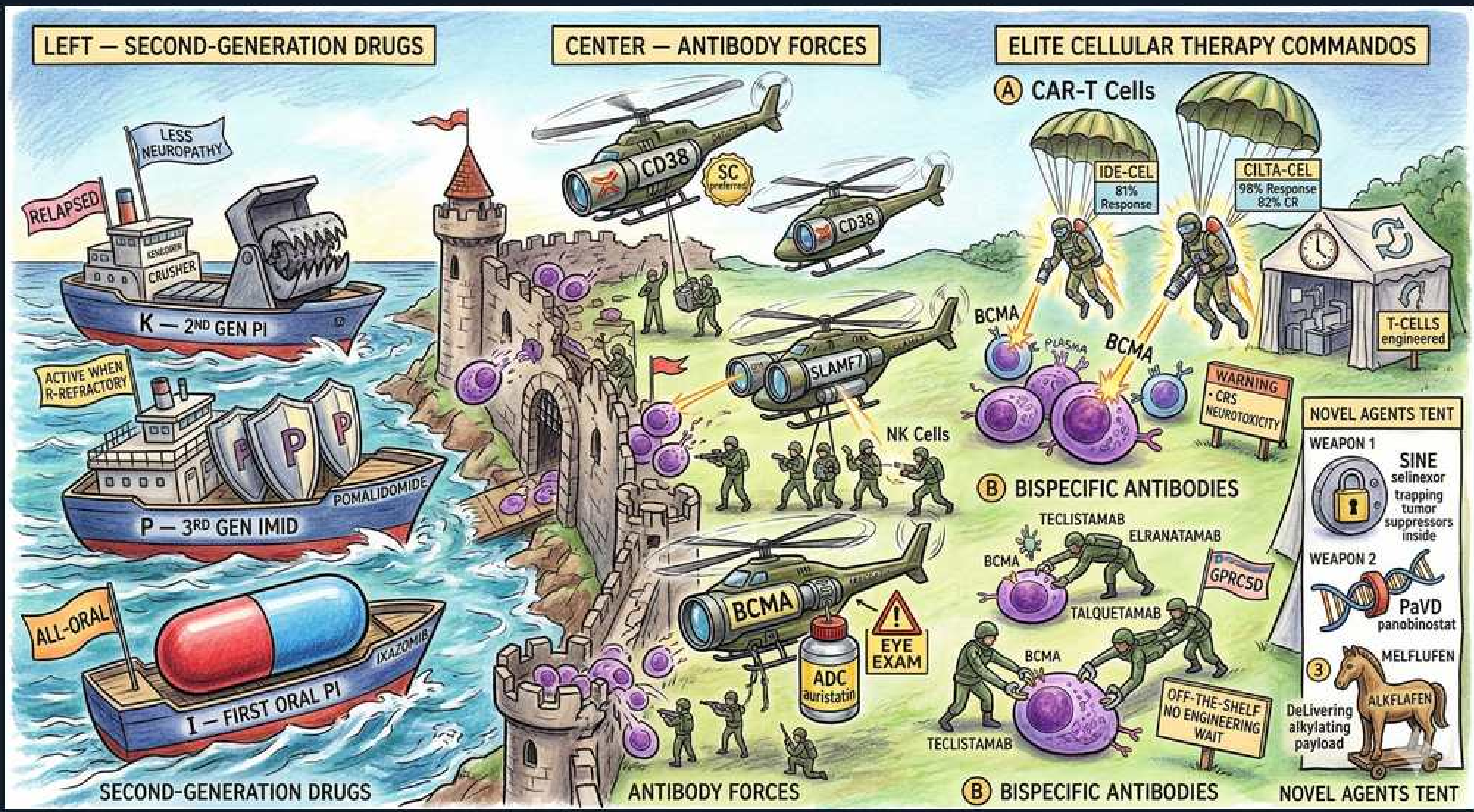
Plasma Cell — Relapsed/Refractory & Novel Immunotherapy



1. 2nd-gen PI carfilzomib

WHAT YOU SEE

Battleship 'K' 2nd-gen PI, less neuropathy

WHAT IT ENCODES

Carfilzomib is a second-generation, IRREVERSIBLE proteasome inhibitor (binds the 20S chymotrypsin-like site covalently) used in relapsed/refractory disease, classically in KRd (carfilzomib-lenalidomide-dexamethasone) or Kd. Its key advantage over bortezomib is that it spares peripheral nerves (little/no neuropathy), but its signature toxicity is CARDIOVASCULAR — heart failure, hypertension, and pulmonary hypertension — plus infusion reactions. Mnemonic: K = the heavy battleship that shells the heart, not the nerves.

BOARD TRAP

WRONG answer: 'switch to carfilzomib to AVOID cardiac toxicity / monitor for worsening neuropathy.' Reversed: carfilzomib's hazard is CARDIOTOXICITY (get baseline echo, control BP, watch for new dyspnea/HF), while it is bortezomib that causes neuropathy. Discriminator: new dyspnea/edema on carfilzomib = cardiac toxicity workup; new stocking-glove numbness points to a bortezomib/thalidomide problem instead.

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2. 3rd-gen IMiD pomalidomide

WHAT YOU SEE

Ship 'P' 3rd-gen IMiD, active when refractory

WHAT IT ENCODES

Pomalidomide is a third-generation immunomodulatory drug (IMiD) that retains activity even after the disease becomes refractory to lenalidomide, making it a relapse workhorse (e.g., Pd, or pomalidomide + bortezomib/dara + dex). Like all IMiDs it requires VTE thromboprophylaxis and is strictly teratogenic (REMS/pregnancy-prevention). Mnemonic: P = the ship that still sails when lenalidomide has run aground.

BOARD TRAP

WRONG answer: 'if a patient is refractory to lenalidomide, all IMiDs are useless.' Pomalidomide works specifically AFTER lenalidomide failure. Discriminator: 'lenalidomide-refractory' on the boards is the cue to choose pomalidomide-based therapy, not to abandon the IMiD class — and don't forget thromboprophylaxis when you start it.

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3. 1st oral PI ixazomib

WHAT YOU SEE

All-oral capsule 'I' first oral PI

WHAT IT ENCODES

Ixazomib is the FIRST ORAL proteasome inhibitor (reversible 20S inhibitor), enabling fully oral relapse regimens such as IRd (ixazomib-lenalidomide-dexamethasone). Its appeal is convenience (no infusion/injection); it has less neuropathy than IV bortezomib, with GI toxicity and thrombocytopenia. Mnemonic: I = the all-oral capsule, the pill you can take at home.

BOARD TRAP

WRONG answer: 'bortezomib is the only orally available proteasome inhibitor.' Bortezomib is given SUBCUTANEOUSLY/IV, not orally — IXAZOMIB is the oral PI. Discriminator: if a question stresses an 'all-oral regimen' for a proteasome inhibitor, the answer is ixazomib (IRd), not bortezomib.

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4. Anti-CD38 daratumumab

WHAT YOU SEE

CD38 helicopters bombarding the castle

WHAT IT ENCODES

Daratumumab is an anti-CD38 monoclonal antibody (CD38 is highly expressed on plasma cells) that kills via ADCC, ADCP, CDC, and direct apoptosis; isatuximab is a related anti-CD38 antibody. It is a backbone of many relapse combinations (DRd, DVd, DPd) and increasingly frontline. Premedicate to reduce infusion reactions. Mnemonic: CD38 helicopters bombard the plasma-cell castle.

BOARD TRAP

WRONG answer: 'an unexpected positive antibody screen / pan-reactive antibody panel before transfusion means a new alloantibody — delay transfusion to identify it.' Anti-CD38 binds CD38 on reagent RBCs causing a FALSE pan-agglutination on the indirect Coombs/antibody screen, and it co-migrates with IgG-kappa on immunofixation (can mimic/mask the M-spike). Discriminator: treat RBCs with DTT (or use a known anti-CD38 history) for type-and-screen; phenotype/genotype the patient BEFORE starting daratumumab.

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5. SLAMF7 elotuzumab

WHAT YOU SEE

SLAMF7 helicopter

WHAT IT ENCODES

Elotuzumab is a monoclonal antibody against SLAMF7 (CS1), expressed on myeloma cells AND on NK cells — so it both tags tumor cells and directly activates NK-mediated ADCC. It has essentially NO single-agent activity and must be combined, typically with lenalidomide-dexamethasone (Elo-Rd) or pomalidomide-dex. Mnemonic: SLAMF7 helicopter only flies in formation (with an IMiD).

BOARD TRAP

WRONG answer: 'use elotuzumab as monotherapy in a frail relapsed patient who can't tolerate combinations.' Elotuzumab is INACTIVE as a single agent — it requires a partner (an IMiD) to work via NK-cell activation. Discriminator: contrast with daratumumab, which DOES have single-agent activity; if the stem demands monotherapy efficacy among antibodies, elotuzumab is the wrong choice.

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6. NK cells

WHAT YOU SEE

NK cell soldiers

WHAT IT ENCODES

Natural killer (NK) cells are the effector arm for much antibody therapy in myeloma: anti-CD38 (daratumumab/isatuximab) and anti-SLAMF7 (elotuzumab) recruit NK cells to perform antibody-dependent cellular cytotoxicity (ADCC). Elotuzumab additionally engages SLAMF7 ON the NK cell to boost its activity directly. Mnemonic: NK-cell soldiers are the ground troops the antibodies call in.

BOARD TRAP

WRONG answer: 'monoclonal antibodies in myeloma work mainly through T cells.' The classic naked antibodies (dara, isa, elo) act largely via NK-cell ADCC/innate effectors — it is the BISPECIFICS and CAR-T that redirect T cells. Discriminator: T-cell engagement (CRS, ICANS) is the fingerprint of bispecifics/CAR-T, whereas naked-antibody toxicity is infusion reactions, not CRS.

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7. BCMA target

WHAT YOU SEE

BCMA flags marking plasma-cell antigens

WHAT IT ENCODES

BCMA (B-cell maturation antigen, TNFRSF17) is the dominant plasma-cell-restricted target in heavily pretreated myeloma and is hit by THREE modalities: CAR-T (ide-cel, cilta-cel), bispecific T-cell engagers (teclistamab, elranatamab), and the ADC belantamab mafodotin. Its near-exclusive expression on plasma cells gives a wide therapeutic window. Mnemonic: BCMA flags mark every plasma-cell target the commandos aim at.

BOARD TRAP

WRONG answer: 'BCMA-directed therapy can only be given once because it permanently eliminates the target.' Sequencing of BCMA agents is an active strategy, though prior BCMA exposure can reduce efficacy of subsequent BCMA therapy (antigen loss/low BCMA). Discriminator: GPRC5D (talquetamab) and CD38 are alternative non-BCMA targets used to overcome BCMA escape.

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8. ADC belantamab

WHAT YOU SEE

ADC mortar + EYE EXAM warning

WHAT IT ENCODES

Belantamab mafodotin is an anti-BCMA antibody-drug conjugate carrying the microtubule toxin MMAF (monomethyl auristatin F). Its signature, dose-limiting toxicity is OCULAR — keratopathy ('microcyst-like epithelial changes'), blurred vision, and dry eye — requiring baseline and serial slit-lamp eye exams (REMS) and dose holds. Mnemonic: the ADC mortar fires the payload but the warning sign says EYE EXAM.

BOARD TRAP

WRONG answer: 'a relapsed myeloma patient on therapy with new blurry vision/keratopathy — this is cataracts or unrelated, continue treatment.' Keratopathy is the hallmark adverse effect of BELANTAMAB and mandates ophthalmology evaluation and dose modification/hold. Discriminator: ocular toxicity points specifically to belantamab (anti-BCMA ADC), not to CAR-T, bispecifics, or proteasome inhibitors.

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9. CAR-T ide-cel/cilta-cel

WHAT YOU SEE

CAR-T paratrooper commandos ide-cel/cilta-cel

WHAT IT ENCODES

Anti-BCMA CAR-T cells — idecabtagene vicleucel (ide-cel) and ciltacabtagene autoleucel (cilta-cel) — are autologous T cells engineered to express a BCMA-targeting chimeric antigen receptor; cilta-cel shows the deepest, most durable responses in triple-class-refractory disease. Requires lymphodepleting chemotherapy before infusion. Mnemonic: the CAR-T paratroopers are elite commandos dropped in for the toughest battles.

BOARD TRAP

WRONG answer: 'CAR-T is an off-the-shelf, immediately available option for rapidly progressing disease.' CAR-T is AUTOLOGOUS and requires apheresis plus weeks of manufacturing — there is a real delay, and patients may need bridging therapy. Discriminator: if the question needs an IMMEDIATELY available T-cell therapy, choose a BISPECIFIC (off-the-shelf), not CAR-T.

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10. CRS / neurotoxicity

WHAT YOU SEE

CRS warning sign near CAR-T cells

WHAT IT ENCODES

T-cell-redirecting therapies (CAR-T and bispecifics) cause CYTOKINE RELEASE SYNDROME (fever, hypotension, hypoxia; IL-6 driven — treat with tocilizumab ± steroids) and NEUROTOXICITY/ICANS (confusion, aphasia, seizures — treated with steroids, not tocilizumab if isolated CNS). They also cause prolonged cytopenias and hypogammaglobulinemia/infection risk. Mnemonic: the CRS warning sign hangs over the T-cell drop zone.

BOARD TRAP

WRONG answer: 'treat ICANS/neurotoxicity with tocilizumab.' Tocilizumab is for CRS; isolated ICANS is treated primarily with CORTICOSTEROIDS (tocilizumab can worsen CNS by raising CNS IL-6). Discriminator: hypotension/fever/hypoxia after CAR-T or bispecific = CRS → tocilizumab; confusion/aphasia/seizure = ICANS → dexamethasone.

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11. Bispecific antibodies

WHAT YOU SEE

Teclistamab/talquetamab/elranatamab bridging T-cell to plasma cell

WHAT IT ENCODES

Bispecific T-cell engagers physically bridge a CD3+ T cell to a myeloma antigen: teclistamab and elranatamab target BCMA, while talquetamab targets GPRC5D. They are OFF-THE-SHELF (no patient-specific manufacturing) and given with step-up dosing to limit CRS. Like CAR-T they cause CRS, ICANS, cytopenias, and infections (hypogammaglobulinemia). Mnemonic: a two-armed antibody grabs a T cell with one hand and the plasma cell with the other.

BOARD TRAP

WRONG answer: 'all bispecifics in myeloma target BCMA, so prior BCMA CAR-T failure rules them all out.' TALQUETAMAB targets GPRC5D (a DIFFERENT antigen), so it remains an option after BCMA-directed failure. Discriminator: talquetamab's unique toxicities — dysgeusia/taste changes, dysphagia, skin rash, and nail changes — flag the GPRC5D target rather than BCMA.

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12. Off-the-shelf advantage

WHAT YOU SEE

'Off-the-shelf no engineering wait' banner

WHAT IT ENCODES

The key practical contrast: BISPECIFICS are ready immediately ('off-the-shelf,' no engineering wait), whereas CAR-T requires apheresis and weeks of manufacturing. For a patient with rapidly progressing disease who cannot wait for CAR-T production, a bispecific is the more feasible choice; CAR-T trades that delay for potentially deeper, one-time durable responses. Mnemonic: the banner reads 'OFF-THE-SHELF — NO ENGINEERING WAIT.'

BOARD TRAP

WRONG answer: 'choose CAR-T for the patient who is decompensating and cannot wait.' The manufacturing lag makes CAR-T unsuitable when time is critical — pick the off-the-shelf BISPECIFIC. Discriminator: 'rapidly progressive / cannot wait weeks' = bispecific; 'wants a single, durable one-time treatment and can tolerate the wait + bridging' = CAR-T.

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13. Selinexor (XPO1)

WHAT YOU SEE

Novel-agents tent weapon 1 selinexor

WHAT IT ENCODES

Selinexor is an oral SINE (Selective Inhibitor of Nuclear Export) that blocks XPO1/exportin-1, trapping tumor-suppressor proteins (p53, etc.) and reducing oncoprotein translation inside the nucleus — used in penta-refractory disease (e.g., XVd: selinexor-bortezomib-dex). Prominent toxicities: nausea/anorexia/weight loss, thrombocytopenia, hyponatremia, fatigue. Mnemonic: Weapon 1 in the novel-agents tent jams the nuclear EXPORT door.

BOARD TRAP

WRONG answer: 'selinexor works by inhibiting the proteasome / is interchangeable with bortezomib.' Its mechanism is XPO1 NUCLEAR EXPORT inhibition, a distinct pathway. Discriminator: significant GI toxicity (nausea, weight loss), thrombocytopenia, and hyponatremia in a heavily-pretreated patient on an oral agent point to selinexor.

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14. Melflufen

WHAT YOU SEE

Novel-agents tent melflufen alkylating payload

WHAT IT ENCODES

Melflufen (melphalan flufenamide) is a peptide-conjugated alkylating prodrug: lipophilic uptake into plasma cells, where aminopeptidases (enriched in myeloma cells) cleave it to release MELPHALAN intracellularly, concentrating DNA-alkylating damage in the tumor. Reserved for relapsed/refractory disease; myelosuppression is the main toxicity. Mnemonic: the melflufen 'smart bomb' delivers melphalan inside the cell.

BOARD TRAP

WRONG answer: 'melflufen is a novel targeted antibody/immunotherapy.' It is fundamentally an ALKYLATOR (a melphalan delivery system), not an immunotherapy. Discriminator: its toxicity profile and mechanism mirror conventional alkylator/melphalan (cytopenias), unlike the CRS/ICANS of T-cell engagers.
